

The Khronon Route: the Blanchet–Skordis Theory Through Two Gates, and a Cubic, One-Scale Khronon That Passes Them

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Abstract

PAPER32 closed the programme’s filtered-khronon completion on cosmology and named one open door: a khronon sector that is itself the dark fluid, as in the theory of Blanchet & Skordis (JCAP 11 (2024) 040). We run that theory through the same two gates, using its own equations, and then repair it. *Moving source (passed)*. Its phantom is a conserved fluid carried by the aether, and a galaxy moving uniformly with its phantom is an exact post-Newtonian solution. The costs are that the phantom is assembled rather than generated, that it follows a change of velocity only inside $r \approx 2GM_b/\Delta u^2$, and that the static halo is stable only for a monotone phantom law. *KiDS external field and σ_8 (failed as built)*. The MOND function reads the gradient of the khronon’s own potential. The published benchmark $\mu^{-1} = 22.3 \text{ Mpc}$ puts 24 times the KiDS-1000 bound on it. The benchmark $\mu^{-1} = 223 \text{ kpc}$ passes, but its mass term ends MOND-like lensing at $\sim 80 \text{ kpc}$, and no μ passes both. Integrating the theory’s own scale-dependent sound speed gives linear $\sigma_8 \approx 0.21\text{--}0.26$. *Diagnosis and repair*. All three cosmological failures come from one term, the quadratic part of the khronon’s kinetic function $\mathcal{K}(\mathcal{Q})$. With $\mathcal{K}$ starting at cubic order the galactic mass term is gone (the static limit is pure AQUAL), and the khronon’s Jeans length stays short at every epoch. For late Jeans lengths of 0.3–3 kpc the repaired theory gives $\sigma_8 = 0.811$, forest-scale power within 2% of ΛCDM , a web field in the MOND function at $\leq 1.5 \times 10^{-7} a_0$, and an early equation of state $w \leq 4 \times 10^{-6}$, while keeping the moving phantom. *The programme’s relation*. Giving the MOND function one scale, $\mathcal{J} = \Lambda j(\mathcal{Y}/\Lambda)$, turns $a_0 = \kappa c \sqrt{G\rho_\Lambda}$ into an identity of the action, with $\kappa = 2\sqrt{8\pi}/(3\beta)$ for β the shape coefficient of j . $\kappa = \frac{1}{2}$ is $\beta = 4\sqrt{8\pi}/3$ and returns $a_0 = 9.35 \times 10^{-11} \text{ m s}^{-2}$ from Planck’s Λ ; κ itself is not derived. Still open: the conserved phantom’s mass budget at Mpc radii, the stability of the GR-recovery tail, and the cubic term’s strong coupling. This is a candidate that passes the gates computed here, not a theory.

1 The route

PAPER32 [Z26d] ended with a specification any completion of $a_0 = \kappa c \sqrt{G\rho_\Lambda}$ must meet. It named as an open door “a khronon sector that itself behaves as dust on FRW, the route of Blanchet & Skordis”. That theory [BS24] is

$$S = \frac{c^3}{16\pi G} \int \sqrt{-g} [R - 2\mathcal{J}(\mathcal{Y}) + 2\mathcal{K}(\mathcal{Q})] + S_m, \quad \mathcal{Q} = c\sqrt{-g^{\mu\nu}\partial_\mu\tau\partial_\nu\tau}, \quad \mathcal{Y} = \frac{A_\mu A^\mu}{c^4}, \quad (1)$$

where $A_\mu = c^2 n^\nu \nabla_\nu n_\mu$ is the acceleration of the khronon’s unit normal n_μ . Their MOND function is $\mathcal{J} = \Lambda - \mathcal{Y} + \frac{2c^2}{3a_0} \mathcal{Y}^{3/2} + \dots$ at low acceleration (their eq. 37), with $\mathcal{J} \rightarrow \Lambda_\infty$ at high acceleration (eq. 40). Their kinetic function is $\mathcal{K} = \mu^2(\mathcal{Q} - 1)^2 + \dots$ (eq. 7), in a DBI form (eq. 87) for cosmology. Their post-Newtonian system (eqs. 19–30) has three parts:

- the unitary-gauge lapse $\Xi = \phi - \dot{\sigma} + \frac{1}{2}|\nabla\sigma|^2$, with $\tau = t + \sigma/c^2$;
- a khronon mass density $4\pi G\rho_\tau = -[\nabla \cdot (\mathcal{J}_\mathcal{Y} \nabla \Xi) + \mu^2 \Xi]$, which is the MOND phantom;
- a khronon velocity $v_\tau = -\nabla\sigma$.

In the static case this reduces to AQUAL with a mass term (eq. 35):

$$\nabla \cdot [f(|\nabla\phi|/a_0)\nabla\phi] + \mu^2\phi = 4\pi G\rho_m, \quad f = 1 + \mathcal{J}_\mathcal{Y}. \quad (2)$$

All numbers below come from committed scripts (lanes BSK1, BSX1, BSX2) run on both a_0 footings where a_0 enters. Each lane has a mutation control that must fail.

2 Gate 1: the moving source — passed (BSK1, 8/8)

In unitary gauge the extra sector depends only on the lapse and its spatial gradient: $\mathcal{Q} = 1/N$ and $\mathcal{Y} = \gamma^{ij}\partial_i \ln N \partial_j \ln N$. That is the shift-free class in which L330 found a frozen phantom. BS24’s own equations show why it is not frozen. The divergence of the momentum equation, combined with the time derivative of the Poisson equation and matter continuity, gives

$$\partial_t \rho_\tau + \nabla \cdot (\rho_\tau v_\tau) = 0.$$

The phantom is therefore a conserved fluid carried by the aether’s flow. About flat space its transport is second order: that is L330’s identity, and BS24’s dispersion $\omega = 0$. About a galaxy it is first order. A galaxy moving uniformly with its phantom is an exact solution. With $\sigma = -V \cdot x + V^2 t/2$ the aether moves with the galaxy, $\Xi = \phi$, and eqs. 27a, 29a and 35 all hold (symbolic, in comoving coordinates). With the aether pinned (the control) the same moving phantom violates continuity. Passing the gate has three costs:

- (i) The phantom is conserved, so it is assembled from the cosmic khronon dust, never generated by baryons.
- (ii) About a static halo the fluid obeys, in the WKB approximation,

$$\omega^2 = \frac{4\pi G \rho_\tau (k^2 f_\theta - \mu^2)}{k^2(1 - f_\theta) + \mu^2}, \quad f_\theta = f + y f'(y) \cos^2 \theta. \quad (3)$$

Its static limit is AQUAL’s linear response. For $k \gg \mu$ the frequency is k -independent, so the phantom follows a change of velocity Δu only inside $r \approx 2GM_b/\Delta u^2$: 1.3 kpc for the Milky Way at 620 km s^{-1} , 860 kpc for a $10^{14} M_\odot$ cluster at 1000 km s^{-1} .

- (iii) The halo is stable only if $0 < f_\theta < 1$, i.e. only if the phantom acceleration increases with g_N . This is the monotone-phantom condition L340 found, reached here from the fluid side. Any tail that recovers GR (their eq. 40) makes the phantom fall. Where it falls the fluid is unstable at rate² $= (2 + 2/n)GM/r^3$, the local orbital rate, whatever the tail’s amplitude. With ν_{RAR} the unstable region is the inner Milky Way, with e-folding times of 1–4 Myr at 1–3 kpc. Placing the fall above $g \sim 10^3 a_0$ confines it to where the phantom is below 10^{-3} of the field.

3 Gate 2 and σ_8 : the theory as built fails (BSX1, 8/8)

The argument of the MOND function is the aether’s acceleration, $A = \nabla \Upsilon$, with $\Upsilon = -4\pi G a^2 \rho_K \Delta_K / (k^2 + k_J^2)$ (their eqs. 61b and 67). This is the khronon’s own potential gradient, filtered above the late Jeans length. The late Jeans wavenumber is $k_J = a\mu$ exactly, as the check confirms. The field is not the web’s Newtonian field; that is the field which closed the programme’s C-H/K construction [Z26d]. We integrate two-fluid growth (baryons plus the khronon, with their DBI $w(a)$, $c_{\text{ad}}^2(a)$ and the scale-dependent sound speed of their eq. 71) from $z = 1000$. The controls reproduce the Λ CDM $\sigma_8 = 0.811$, and the Λ CDM web field at $z = 0.25$ comes out at $0.0134 a_0$ against an independent mock’s 0.0150. A pure-dust khronon returns $\sigma_8 = 0.82\text{--}0.85$, so the suppression below is the sound speed alone.

benchmark (their Fig. 1)	MOND-function field at KiDS lenses	vs bound 7.2×10^{-5}	r_C (Milky Way)	linear σ_8
A: $\mu^{-1} = 22.3 \text{ Mpc}$, $\lambda_D = 1$	$1.7 \times 10^{-3} a_0$	$\times 24$, fails	1.7 Mpc	0.21
B: $\mu^{-1} = 223 \text{ kpc}$, $\lambda_D = 30$	$4.0 \times 10^{-5} a_0$	$\times 0.55$, passes	0.08 Mpc	0.26

Across $\mu^{-1} = 0.03\text{--}100 \text{ Mpc}$ (with $\lambda_D = 1$ and 30), KiDS alone leaves no μ . The external-field bound needs $\mu^{-1} \lesssim 0.3 \text{ Mpc}$, while MOND-like lensing to $\sim 1 \text{ Mpc}$ needs $r_C = (r_M/\mu^2)^{1/3} \gtrsim 1 \text{ Mpc}$ (their eq. 38), i.e. $\mu^{-1} \gtrsim 10 \text{ Mpc}$. At the same time σ_8 lies between 0.16 and 0.49 at every scanned point. Misteale, McGaugh & Hossenfelder [MMH23] confronted the full static profile of AeST’s identical mass term with KiDS. BS24 checked $w(a)$ against GDM constraints and left the scale-dependent sound speed “for a future investigation”; it is integrated here. A parallel lane in the same folder (BSK2) finds the same suppression on a Λ CDM background. BSK3 finds that khronon dust at the cosmic abundance under-supplies KiDS’s Mpc-scale flux ($\Delta\chi^2 = +374/+364$ on all points).

4 The diagnosis: one term

In BS24 the single constant μ of the quadratic term sets three things:

- the galactic mass term $\mu^2 \phi$, which ends MOND at r_C ;
- the late Jeans length $1/\mu$, which sets σ_8 ;
- the filter that hides the web’s field from the MOND function, which sets KiDS.

Because the comoving Jeans length is $(1+z)/\mu$, a Jeans length short enough today is a thousand times longer at recombination. BS24 themselves note (their sec. 4.3.2, eq. 86) that for $\mathcal{K}$ starting at cubic order the kinetic term drops out of the Newtonian limit.

5 The repair: a cubic, one-scale khronon (BSX2, 3/3)

$$\mathcal{K}(\mathcal{Q}) = \frac{2M^2}{3}(\mathcal{Q} - 1)^3, \quad \mathcal{J}(\mathcal{Y}) = \Lambda j(\mathcal{Y}/\Lambda), \quad j(u) = 1 - u + \beta u^{3/2} + \dots \quad (4)$$

Galaxies. The cubic term contributes $c^2 \mathcal{K}_{\mathcal{Q}} = 2M^2 \Xi^2 / c^2$, a 1PN correction. The static limit is therefore pure AQUAL with no mass term and no r_C (symbolic; the quadratic control carries BS24's $-2\mu^2 \Xi$). BS24's moving phantom (Sec. 2) is unchanged, because the PN phantom dynamics never involved $\mathcal{K}$ beyond the mass term. *Cosmology.* Their eqs. 82–85 with $n = 2$ give $q = \mathcal{Q} - 1 = (I_0/2M^2 a^3)^{1/2}$, $w = q/(3 + 2q)$ and $c_{\text{ad}}^2 = q/[2(1 + q)]$. The comoving Jeans length is $\propto (1 + z)^{1/4}$, only 5.6 times longer at $z = 1000$ than today.

late Jeans length	σ_8	forest power ($z=3$, $k = 1\text{--}10 h/\text{Mpc}$)	web field in MOND fn.	early w	all gates
0.3 / 1 / 3 kpc	0.811	$\geq 1.000 / 0.998 / 0.981$	$\leq 1.5 \times 10^{-7} a_0$	$\leq 3.8 \times 10^{-6}$	pass
10 kpc	0.811	0.82	$1.6 \times 10^{-6} a_0$	4×10^{-5}	forest fails
100 kpc / 1 Mpc	0.806 / 0.597	≤ 0.035	$4 \times 10^{-5} / 2 \times 10^{-4} a_0$	$\geq 4 \times 10^{-3}$	fail

The criteria are σ_8 within 5% of 0.811, forest-scale matter power within 10%, the KiDS bound on both footings, and early $w \leq 0.0164$ at $a = 10^{-5}\text{--}10^{-3}$. The repaired khronon passes all four for late Jeans lengths of a few kpc or less. It clusters as cold dark matter on every linear scale, and its short Jeans length removes the web's field from the MOND function. The control restores the quadratic term at the same late Jeans length. It then has a stiff early epoch ($w \rightarrow 0.99$) and loses the forest (power 0.08 of ΛCDM).

6 The programme's relation, inside the action

$\mathcal{J}(0) = \Lambda$ is the cosmological constant that cosmology measures, and the $\mathcal{Y}^{3/2}$ coefficient is $2c^2/(3a_0)$ (their eq. 37). With one scale in $\mathcal{J}$,

$$a_0 = \frac{2c^2 \sqrt{\Lambda}}{3\beta} = \kappa c \sqrt{G\rho_\Lambda}, \quad \kappa = \frac{2\sqrt{8\pi}}{3\beta}, \quad \rho_\Lambda = \frac{\Lambda c^2}{8\pi G} \quad (5)$$

(symbolic). So $\kappa = \frac{1}{2}$ is $\beta = 4\sqrt{8\pi}/3 = 6.684$, and Planck's Λ ($h = 0.6736$, $\Omega_\Lambda = 0.6847$) then returns $a_0 = 9.35 \times 10^{-11} \text{ m s}^{-2}$, the canonical footing. The relation stops being a coincidence between two constants: a_0 and the vacuum energy become two coefficients of one function. The programme's earlier principle that there is no independent a_0 [Z26b] becomes a property of a covariant action whose phantom passes the moving-source gate. The coefficient remains a shape number and is not derived, consistent with the zero-mode theorem [Z26a].

7 What is still open

- (i) *The mass budget.* A conserved phantom must be assembled from the khronon dust that falls in; it cannot be generated. The parallel lane BSK3 reports that khronon dust at the cosmic abundance under-supplies KiDS's Mpc-scale flux. This lane does not compute the budget. Whether nonlinear MOND infall, which is faster than ΛCDM 's, closes the gap is the construction's next gate.
- (ii) *The GR-recovery tail.* It is unstable where it falls (Sec. 2). Placing the fall above $\sim 10^3 a_0$ hides the instability, but the ephemeris bound on the monotone stretch must then be checked with the programme's Solar-System machinery.
- (iii) *Strong coupling.* The cubic kinetic term makes $\mathcal{K}_{\mathcal{Q}\mathcal{Q}} \rightarrow 0$ at $\mathcal{Q} \rightarrow 1$, which is not analysed here. BS24's Minkowski-space mode analysis would need redoing without the quadratic term.
- (iv) *The successor theory.* Blanchet & Skordis's Khronon-Tensor theory [BS25] is not examined.

8 Reproducibility

All lanes are in <https://github.com/carlzimmerman/zimmerman-formula> under `real_research/bs_khronon_2026/`. They run in about a minute each and need no survey data files.

lane	script	checks	mutation control (must fail)
BSK1	<code>BSK1_moving_source_gate.py</code>	8/8	aether pinned: moving solution and fluid response fail

lane	script	checks	mutation control (must fail)
BSX1	BSX1_kids_external_field_gate.py	8/8	khronon made pure dust: the σ_8 and field suppressions vanish
BSX2	BSX2_cubic_one_scale_khronon.py	3/3	quadratic term restored at the same Jeans length: mass term returns, forest and early w fail

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